

DEV_IMPL_GUIDE - DD: OSR-RMA-01-FRAMEWORK — Equipment Swap & RMA Workflow Framework

1. Purpose

This guide explains how to implement DD_OSR-RMA-01-FRAMEWORK-v1.0.md as executable SOPHIX behavior.

Use it with:

- DD_OSR-RMA-01-FRAMEWORK-v1.0.md
- DD_SUB-WF-FRAMEWORK-01-Subscription_Workflow_Framework-v1.0.md when this DD participates in subscription workflow operations.
- DD_SUB-WF-FRAMEWORK-01-WORKERS-v1.0.md when this DD references worker topics.

2. Runtime Actors

Runtime component	Responsibility
API pod / module service	Receives requests, validates, loads authoritative state, starts commands.
Owning module database	Stores tables/configuration owned by this DD.
Reference modules	Provide data through approved APIs/client wrappers.
Camunda	Runs BPMN process when this DD is workflow-driven.
Worker apps	Execute logical worker topics.
Drools	Evaluates packages when rules are declared.
Kafka/outbox	Publishes success/failure and integration events.

3. Before Implementation

Confirm these contracts from the DD:

Contract	Value
Endpoint scan reference	DELETE /engine-rest/process-instance/{id}
BPMN/process key	configured BPMN process
Drools package	configured Drools package
Tables	equipment_swap_request, equipment_swap_request_history, vendor_rma_stub
Events	EquipmentInstanceLifecycleTransition

4. Execution Flow

1. Apply migrations and seed configuration/catalog rows.
2. Implement the module service/repository layer around the tables and invariants.
3. Implement public/internal APIs from the DD.
4. Implement rule evaluation if the DD defines Drools packages.
5. Implement event production/consumption through outbox where the DD defines events.
6. Add integration tests for table constraints, API validation, and event payloads.

5. API And Command Handling

Endpoints found in the DD:

- DELETE /engine-rest/process-instance/{id}
- GET /api/swap-requests/{swap_request_id}

- GET /api/swap-requests/{swap_request_id}/history
- GET /api/swap-requests?operator=WIK&accountId=acct_01HZ_NJERI
- GET /api/swap-requests?operator=WIK¤tState=FAILED&since=2026-03-01
- GET /api/swap-requests?operator=WIK&recoveryContractorId=contractor_ke_nairobi_a
- GET /api/swap-requests?operator=WIK&swapKind=HFC_FIELD¤tState=AWAITING_SLOT
- PATCH /api/contractor-slot-commitments/{id}
- PATCH /api/homepasses/{id}/network-path
- PATCH /api/homepasses/{id}/network-path/...
- PATCH /api/homepasses/{id}/network-path/ports
- PATCH /api/vendor-rma-stubs/{id}
- PATCH /api/vendor-rma-stubs/{stub_id}
- PATCH /api/work-orders/{id}
- POST /api/billable-events
- POST /api/contractor-availability
- POST /api/contractor-slot-commitments
- POST /api/equipment-instances/inst_01HZ_CM_NJERI_OLD/lifecycle-events
- POST /api/equipment-instances/{id}/lifecycle-events
- POST /api/equipment-instances/{source_instance_id}/lifecycle-events
- POST /api/equipment-instances/{target_instance_id}/lifecycle-events
- POST /api/stock-movements
- POST /api/swap-requests/...
- POST /api/swap-requests/eqp
- POST /api/swap-requests/{id}/advance
- POST /api/swap-requests/{id}/cancel
- POST /api/swap-requests/{swap_request_id}/advance
- POST /api/swap-requests/{swap_request_id}/cancel
- POST /api/vendor-rma-stubs
- POST /api/work-orders
- POST /engine-rest/external-task/extt_01HZ_inst_state/complete
- POST /engine-rest/external-task/fetchAndLock

Implementation rules:

- Resolve actor user and role from auth context, not from public request body.
- Check idempotency before inserting a new operation or creating a side effect.
- Return the existing operation/result for the same idempotency key and same request hash.
- Return 409 IDEMPOTENCY_KEY_REUSED for the same key with a different request hash.
- Use transactions or unique constraints for concurrency-sensitive creation.

6. Data Loading

Load data in this order:

1. Authoritative aggregate state from the owning module.
2. Operator/module config from this DD's config tables.

3. Catalog/reference data from approved APIs or client wrappers.
4. Current in-flight operation state from the framework, where relevant.

Do not read current mutable state from cache for state-changing operations.

7. Drools Configuration

Packages found:

- No Drools package explicitly declared.

For each package:

- Define fact classes that mirror the request, current state, config, and reference data.
- Keep rule output as explicit decision fields.
- Unit-test accept and reject cases from the DD.
- BPMN should re-check critical rules after start if state may have changed.

8. BPMN And Worker Topics

Worker topics found:

- assign-contractor
- bil01-charge-pre-authorize
- bpmn-cancel-requested
- contractor-slot-release
- contractor-slot-reserve
- contractor-stock-route
- customer-appointment-accepted
- customer-appointment-rejected
- dispatch-agent-wik-01
- eligibility-check
- field-techs
- finance-warranty-approvers
- homepass-network-path-update
- instance-lookup
- instance-state-transition
- nms-remote-bind-stub
- nms-remote-completed
- nms-remote-unassign-stub
- noc-agent-wik-01
- noc-confirm-remote-success
- noc-team
- notification-send
- payment-confirmed
- payment-timeout
- provisioning-adapter
- remote-attempt-timeout
- routing-decision-on-recovery

- routing-decision-recovered-instance-routes-to-recovering-contractor
- serial-rebind-broadhub-stub
- serial-unbind-broadhub-stub
- swap-dispatch-slot-selection
- swap-dispatchers
- swap-eligibility
- swap-request-history-append
- system-bpmn-engine
- tech-confirm-defect
- tech-confirm-recovery-or-eqr
- vendor-rma-coordinator-batch-ship
- vendor-rma-coordinators
- warranty-approver-decision
- wik-internal-tech-team
- wo-finalized

Implementation rules:

- Model service tasks with `camunda:type="external"` and the exact topic string.
- Worker handler must validate required variables before calling downstream services.
- Store outputs as process variables with stable names.
- Use message correlation keys exactly as specified by the DD.
- Add compensation paths for each external side effect.

Notification worker rule:

- `notification-send` must pass only `recipientKind`, `recipientRef`, `eventKind`, `notificationVars`, and `correlationRef`.
- Do not pass `templateId`, `channel`, or `language` from OSR-RMA. NOT-01/ICN-01 owns template selection, channel choice, language, fallback, and delivery retry policy.

9. Events

Events found:

- `EquipmentInstanceLifecycleTransition`

Event rules:

- Emit success only after authoritative commit.
- Emit rejected/failure event only after compensation/final failure decision.
- Use the outbox table/dispatcher; do not publish directly inside the transaction boundary unless the platform standard allows it.

10. Smoke Tests

Add tests for:

- Happy path.
- Invalid role or trigger.
- Invalid state.
- Missing config/catalog row.
- Dependency failure.

- Retry/idempotency.
- Event payload correctness.
- Worker topic failure and compensation.

11. Common Mistakes

- Treating worker-topic names as shell commands.
- Querying another module's database directly.
- Reading current state from cache in a state-changing flow.
- Emitting success before final commit.
- Hiding manual recovery paths for partial external side effects.
- Creating one deployment per topic when a consolidated worker app would be easier to operate.